LLM-Informed Bayesian Priors for Reducing Experimental Cost in Behavioral Economics

Project Check-in Report

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1 Problem Definition

We study Bayesian inference over human behavioral parameters in economic experiments, using large language model (LLM) simulations as an informative prior. Let $\theta = (\alpha, \beta)$ parametrize a Beta likelihood $p(y \mid \theta) = \text{Beta}(y; \alpha, \beta)$ for sharing ratios $y \in (0, 1)$ in a dictator game. Given n human observations $\mathbf{y} = \{y_1, \ldots, y_n\}$, the posterior is

$$p(\theta \mid \mathbf{y}) \;\propto\; p(\mathbf{y} \mid \theta) \cdot p(\theta). \tag{1}$$

Our primary quantity of interest is the **sample size reduction ratio** $\rho = n^*_{\text{LLM}}/n^*_{\text{flat}}$, where n^* is the minimum number of human observations to reach a target posterior precision.

Data source adjustment. Our proposal planned to use dictator game data from Horton et al. [2023]. Upon investigation, Horton's "dictator games" replicate Charness and Rabin [2002]—binary allocation choices between fixed payoff pairs, not continuous sharing ratios. This format is incompatible with our Beta likelihood model. We therefore use individual-level dictator game data from von Blanckenburg et al. [2023] (137 dictators, endowment €10, giving and taking variants), where sharing ratio $y = \text{amount_given}/10 \in [0, 1]$ directly fits our model.

2 Baseline: Flat Prior + MCMC

Method. We use a flat (uniform) prior $p(\alpha, \beta) \propto 1$ on $(0, 100)^2$ and sample the posterior via Metropolis–Hastings (MH) in log-transformed space $(\log \alpha, \log \beta)$ with Gaussian proposals ($\sigma = 0.3$), 15,000 samples after a 3,000-step burn-in.

Results. Table 1 reports posterior summaries for the mean sharing ratio $\mu = \alpha/(\alpha + \beta)$ across sample sizes $n \in \{5, 10, 20, 50, 90\}$ drawn from the training set (96 observations; 30% held out for calibration). As expected, posterior variance decreases monotonically with n. At $n = 90$, the posterior mean $\hat{\mu} = 0.323$ with 95% credible interval width ≈ 0.11. MH acceptance rates range from 0.22 ($n = 90$) to 0.58 ($n = 10$), indicating reasonable chain mixing.

n	$\hat{\mu}_{\text{flat}}$	Std(μ)	Var(μ) $\times 10^3$	Accept rate
5	0.408	0.072	5.16	0.54
10	0.240	0.069	4.81	0.58
20	0.263	0.054	2.88	0.48
50	0.270	0.034	1.15	0.32
90	0.323	0.028	0.77	0.22

Table 1: Baseline (flat prior + MCMC) posterior summaries for μ.

3 Frontier: LLM Prior + SVGD

LLM prior construction. We queried GPT-4o (`gpt-4o`, temperature=1.0) with a standard dictator game prompt: *"You receive €10. How much do you give to an anonymous stranger?"* We collected $M = 500$ simulated sharing ratios across 10 batches. The LLM prior distribution has mean $\bar{y}_{\text{LLM}} = 0.299$ and standard deviation 0.231, compared to the human data mean of 0.361. The prior enters (1) as a power prior:

$$p_{\text{LLM}}(\alpha, \beta) \;\propto\; \prod_{j=1}^{M} \text{Beta}(\tilde{y}_j;\, \alpha, \beta)^{w/M}, \tag{2}$$

where $w = 20$ is the effective weight (equivalent to 20 human observations).

SVGD implementation. We use Stein Variational Gradient Descent with 100 particles in log-transformed space, Adam optimizer (step size 0.05), RBF kernel with median bandwidth heuristic, and 2,000 iterations. Particles are initialized via method-of-moments from the data.

Comparison to baseline. Table 2 and Figure 1 compare posterior variance across sample sizes. The LLM prior consistently reduces posterior variance by a factor of 1.3–2.4× relative to the flat prior at every sample size. Setting the target precision at the midpoint between $\text{Var}(\mu)$ at $n = 5$ and $n = 90$ (flat prior), we obtain:

$$n^*_{\text{flat}} \approx 19.2, \quad n^*_{\text{LLM}} \approx 5.0, \quad \rho = \frac{n^*_{\text{LLM}}}{n^*_{\text{flat}}} \approx 0.26. \tag{3}$$

The LLM prior reduces the required human sample size by approximately **74%**.

n	$\text{Var}(\mu)_{\text{flat}} \times 10^3$	$\text{Var}(\mu)_{\text{LLM}} \times 10^3$	Variance ratio
5	5.16	2.17	2.38×
10	4.81	1.84	2.62×
20	2.88	1.41	2.04×
50	1.15	0.81	1.42×
90	0.77	0.61	1.27×

Table 2: Posterior variance comparison: flat prior vs. LLM prior (MCMC).

4 Preliminary Failure Mode Analysis

We test robustness to prior misspecification by artificially shifting the LLM pseudo-observations by $\delta \in \{-2, -1, 0, +1, +2\}$ standard deviations and measuring empirical coverage of 95% posterior predictive intervals on the held-out test set ($n_{\text{test}} = 41$).

Key finding: The LLM prior is most harmful when moderately misspecified in the upward direction ($\delta = +1$, prior mean ≈ 0.53 vs. data mean 0.36). Extreme misspecification ($|\delta| = 2$) is less harmful because the data overwhelms the prior. This asymmetry reflects the fact that GPT-4o's unshifted prior ($\bar{y} = 0.299$) lies slightly below the human data mean, so upward shifts move it further from truth while downward shifts or extreme shifts are neutralized by the data.

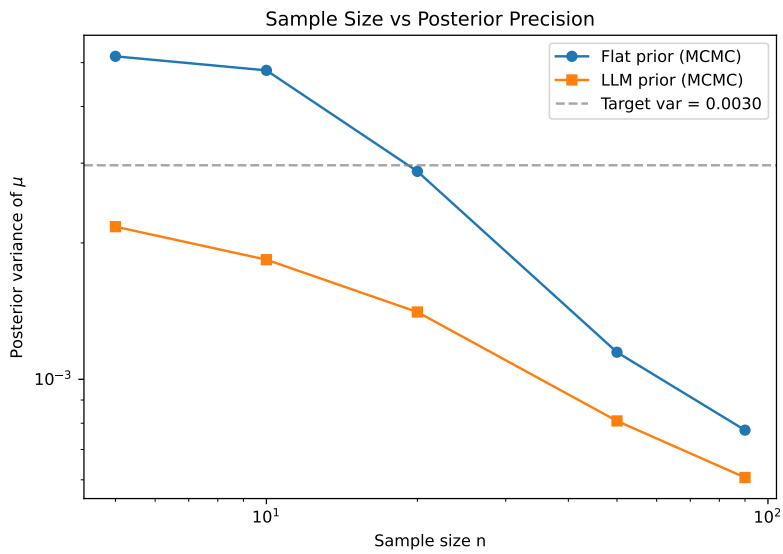

Figure 1: Posterior variance of μ vs. sample size n for flat and LLM priors. The LLM prior crosses the target precision threshold at $n \approx 5$, while the flat prior requires $n \approx 19$.

Prior shift δ	Pseudo-obs mean	Coverage	Nominal
-2	≈ 0.10	0.976	0.95
-1	≈ 0.20	0.976	0.95
0 (original)	0.299	0.976	0.95
$+1$	≈ 0.53	**0.829**	0.95
$+2$	≈ 0.76	0.976	0.95

Table 3: Coverage under prior misspecification. Upward shift of +1 std degrades coverage to 83%.

5 Evaluation Plan & Remaining Work

Completed.

- Data preparation (137 dictator game observations, train/test split)
- LLM prior elicitation (500 GPT-4o pseudo-observations)
- Baseline: flat prior + MCMC (Metropolis–Hastings)
- Frontier: LLM prior + SVGD (Stein Variational Gradient Descent)
- Sample size reduction ratio $\rho = 0.26$
- Preliminary failure mode analysis (prior misspecification)

Remaining for final report (due April 27).

1. **Extended calibration:** Coverage analysis at multiple significance levels and sample sizes; calibration plots (expected vs. observed coverage).

2. **Compute–accuracy tradeoff:** Wall-clock time comparison and effective sample size for SVGD vs. MCMC at matched posterior precision.
3. **Game variant comparison:** Run the full pipeline on the "taking game" variant (68 observations) to test whether the LLM prior generalizes across framing conditions.
4. **Finer failure mode analysis:** Denser shift grid ($\delta \in [-3, +3]$ in steps of 0.5); coverage vs. prior weight w sensitivity.
5. **Reproducibility:** Fixed seeds, `environment.yml`, README with exact reproduction commands.

Code repository. All code is available at: https://github.gatech.edu/bfeng66/cse8803-iuq-project (private, collaborators `pchen402` and `psi6` added).

References

Gary Charness and Matthew Rabin. Understanding social preferences with simple tests. *Quarterly Journal of Economics*, 117(3):817–869, 2002.

John J. Horton, Apostolos Filippas, and Benjamin S. Manning. Large language models as simulated economic agents: What can we learn from homo silicus? *arXiv preprint arXiv:2301.07543*, 2023.

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